

BOURBAKI SEMINAR
32nd year, 1979/80, no. 543
November 1979

**THE FUNDAMENTAL GROUP OF THE COMPLEMENT
OF A PLANE CURVE HAVING ONLY ORDINARY
DOUBLE POINTS IS ABELIAN**

[after W. FULTON]

by Pierre DELIGNE

Introduction

THEOREM 1.- Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a plane curve whose only singularities are double points with distinct tangents. Then the fundamental group of $\mathbb{P}^2(\mathbb{C}) - C$ is abelian.

This theorem was stated by Zariski in 1929 ([8]). See also [9], Ch. VIII. His proof used Severi's theorem, according to which the space of irreducible plane curves of a given degree having a given number of double points is irreducible (Vorlesungen über algebraische Geometrie, 1921, Anhang F). It is still unknown whether Severi's assertion is true or false.

The π_1 considered in the theorem is the genuine fundamental group, the topologists' one. If $\mathbb{C}$ is replaced by an arbitrary algebraically closed field k of characteristic 0, the same statement holds for the algebraic fundamental group, the one classifying étale coverings: the Lefschetz principle reduces us to assuming $k = \mathbb{C}$, in which case the algebraic fundamental group is the profinite completion of the genuine fundamental group. In arbitrary characteristic one has the variant:

THEOREM 2 (Fulton [3]).- Let k be an algebraically closed field and, over k , let $C \subset \mathbb{P}^2$ be a plane curve whose only singularities are double points with distinct tangents. Then the tame fundamental group of $\mathbb{P}^2 - C$ is abelian.

In other words, coverings (in the sense of 0.2) of $\mathbb{P}^2$, ramified only along C and tamely ramified, are abelian. Recall that tameness of the ramification is tested after base change from $\mathbb{P}^2$ to its localizations at the generic points of the irreducible components of C . These localizations are traits (spectra of discrete valuation rings).

Suppose k has characteristic $p > 0$. If C is as in Theorem 2, there always exists in relative projective space $\mathbb{P}^2$ over $\text{Spec}(W(k))$ a relative normal-crossings divisor $\widetilde{C}$ whose special fiber is C (see, for example, Mumford, Appendix to Ch. VIII of [9]). This makes it possible to deduce Theorem 2 from Theorem 1 by a specialization argument.

Once one knows that $\pi_1(\mathbb{P}^2(\mathbb{C}) - C)$ is abelian, homological arguments immediately give its structure: if C is the union of irreducible curves of degrees $d_1, \dots, d_r$, then $\pi_1(\mathbb{P}^2(\mathbb{C}) - C)$ is the quotient of $\mathbb{Z}^r$ by the subgroup generated by the element $(d_1, \dots, d_r)$.

In his proof of Theorem 2, Fulton uses, on the one hand, methods introduced by Abhyankar [1] to prove the special case of Theorem 2 in which the irreducible components of C are assumed smooth - methods very clearly presented here by Serre ([7]) - and, on the other hand, Theorem 3 below. An analysis of the proof then enabled the expositor to give a variant yielding Theorem 1.

Abhyankar and Fulton both use instances of the principle that, in algebraic geometry, what is very mobile is connected. Let $C \subset \mathbb{P}^2$ be as in Theorem 2, let D be an irreducible component of C , and let $f: X \rightarrow \mathbb{P}^2$ be a covering ramified only along C and tamely ramified. Since $D \subset \mathbb{P}^2$ is very mobile, $f^{-1}(D)$ is connected. More precisely, the divisor $D \subset \mathbb{P}^2$ is ample, hence the divisor $f^{-1}(D) \subset X$ is also ample because f is finite, and one applies Bertini. By a local study of f , Abhyankar shows that if D is smooth then $(f^{-1}(D))_{\text{red}}$ is smooth as well, and concludes that $f^{-1}(D)$ is irreducible.

In the general case, let $g: \tilde{D} \rightarrow D$ be the normalization of D . The same local study shows that the reduced fiber product $(X \times_{\mathbb{P}^2} \tilde{D})_{\text{red}}$ is smooth. It is therefore the normalization of $f^{-1}(D)$. If it can be shown to be connected, it follows that $f^{-1}(D)$ is irreducible, and one can repeat the end of Abhyankar's proof as presented in [7].

Recall the definition of the fiber product: if $h: X \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2$ is the product of the maps $f: X \rightarrow \mathbb{P}^2$ and $g: \tilde{D} \rightarrow \mathbb{P}^2$, then the fiber product $X \times_{\mathbb{P}^2} \tilde{D}$ is the inverse image $h^{-1}(\Delta)$ of the diagonal. Since $\mathbb{P}^2$ has many automorphisms, the diagonal $\Delta \subset \mathbb{P}^2 \times \mathbb{P}^2$ is very mobile, and $h^{-1}(\Delta)$ is connected. This completes the proof. More precisely, one applies to h the special case $r = 2, m = 2$ of the following theorem:

THEOREM 3 (W. Fulton and J. Hansen [4]).- Let $P = (\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P , let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be a complete irreducible variety, let $h: Z \rightarrow P$, let $n = \dim h(Z)$, and let $c = (r - 1)m$ be the codimension of Δ in P . If $n - c \geq 1$, then $h^{-1}(\Delta)$ is connected.

This theorem has many other applications, for which I refer to [4] and to the article of T. Gaffney and R. Lazarsfeld [5].

At the end of this introduction, we explain by an example the method used to transpose Fulton's proof of Theorem 2 into a proof

of Theorem 1. The example is this: how to transpose Abhyankar's arguments to prove Theorem 1 in the special case where C is smooth (hence irreducible).

Let U be a tubular neighborhood of C in $\mathbb{P}^2(\mathbb{C})$, let $U^* = U - C$, and let $u \in U^*$. For every covering $f : X \rightarrow \mathbb{P}^2(\mathbb{C})$ ramified only along C , one has

$$(f^{-1}(C) \text{ connected}) \Leftrightarrow (f^{-1}(U) \text{ connected}) \Leftrightarrow (f^{-1}(U^*) \text{ connected}).$$

The key result - that for every covering X the curve $f^{-1}(C)$ is connected - is therefore equivalent to the following: for every finite quotient H of $\pi_1(\mathbb{P}^2(\mathbb{C}) - C, u)$, the composite map

$$\pi_1(U^*, u) \xrightarrow{j^*} \pi_1(\mathbb{P}^2(\mathbb{C}) - C, u) \rightarrow H$$

is surjective. A Morse-theoretic argument proves more: the map $j^* : \pi_1(U^*, u) \rightarrow \pi_1(\mathbb{P}^2(\mathbb{C}) - C, u)$ is surjective.

After choosing a Riemannian metric on $\mathbb{P}^2(\mathbb{C})$, for $\varepsilon > 0$ sufficiently small one may take U to be the image under the exponential map $\exp$ of the subbundle of balls of radius ε in the normal bundle of C in $\mathbb{P}^2(\mathbb{C})$. For ε sufficiently small, $\exp$ identifies U^* with an oriented punctured-disk bundle over C , giving an exact sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, u) \rightarrow \pi_1(C, \bar{u}) \rightarrow 1$$

($\bar{u}$ is the image of u). The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop γ going once around C in the positive direction.

Since j^* is surjective, γ also defines a central element of $\pi_1(\mathbb{P}^2(\mathbb{C}) - C, u)$. The quotient of $\pi_1(\mathbb{P}^2(\mathbb{C}) - C, u)$ by the normal subgroup generated by γ (in this case, the cyclic subgroup generated by γ) is $\pi_1(\mathbb{P}^2(\mathbb{C}), u) = \{e\}$, and the special case of the theorem under consideration follows.

0. Terminology

0.1. Connected means connected and nonempty. The same convention applies to irreducible.

0.2. A covering of a scheme S , assumed normal and connected, is a morphism $f : T \rightarrow S$ whose source T is normal and connected and whose restriction to a dense open subset of T is étale.

0.3. Throughout the rest of this exposition we work over $\mathbb{C}$. We shall write simply $\mathbb{P}^n$ for $\mathbb{P}^n(\mathbb{C})$. All schemes considered are assumed separated.

1. Theorems of Lefschetz and Bertini type

The special case of 1.1 below in which $Z \subset \mathbb{P}^n$ is the complement of a hypersurface is proved by Hamm and Lê [6], using Morse theory. Our aim in this section is to deduce from it the variant 1.6 of Theorem 3.

Assertion 1.1.- Let $Z \subset \mathbb{P}^n$ be a nonempty Zariski-open subset. If H is a general hyperplane, the natural map $\pi_i(Z \cap H) \rightarrow \pi_i(Z)$ is bijective for $i < n - 1$ and surjective for $i = n - 1$.

Assertion 1.1 is stated in [2], but Cheniot's proof covers only the case in which the complement of Z has codimension ≥ 2 . Under this hypothesis, Z and $Z \cap H$ are simply connected, and it is harmless to work, as he does, with the H_i rather than the π_i . The case $i = 1$ can also be deduced from [6] or [2 bis].

Recall that the adjective "general," qualifying an element H of an irreducible algebraic variety S (here the dual projective space $\mathbb{P}^{n\vee}$), is a quantifier. It means that there exists a nonempty (hence dense) Zariski-open subset S' of S such that for every H in S' ,

1.2.- Let $Y \subset \mathbb{P}^n \times \mathbb{P}^{n\vee}$ be the space of pairs (z, H) such that $z \in Z$ and $z \in H$. The isotopy theorem, applied to the second projection $\text{pr}_2 : Y \rightarrow \mathbb{P}^{n\vee}$, gives a nonempty Zariski-open subset U of $\mathbb{P}^{n\vee}$ over which Y is a topologically locally trivial fiber space. It is enough to take U to be the set of hyperplanes transverse to the strata of an algebraic Whitney stratification of the complement of Z in $\mathbb{P}^n$. Locally over U , after choosing a local trivialization of this fiber space, the inclusion $Z \cap H = \text{pr}_2^{-1}(\{H\})$ into Z appears as a continuously H -dependent map from a fixed space into Z . Thus the conclusion of 1.1 can hold for one H in U only if it holds for all of them, and 1.1 is equivalent to saying that it holds for every H in U .

Suppose that Z depends on parameters: take an irreducible scheme S and an open subset Z of $S \times \mathbb{P}^n$; set $\text{pr}_2 \text{pr}_1^{-1}(\{s\}) = Z(s)$, and assume every $Z(s)$ is nonempty. Let $Y \subset S \times \mathbb{P}^n \times \mathbb{P}^{n\vee}$ be the space of triples (s, z, H) such that $z \in Z(s)$ and $z \in H$. The isotopy theorem, applied to the projection of Y onto $S \times \mathbb{P}^{n\vee}$ (respectively, of Z onto S), gives a nonempty Zariski-open subset U of $S \times \mathbb{P}^{n\vee}$ (respectively V of S) over which Y (respectively Z) is a topologically locally trivial fiber space.

As above, if $s \in V$ and $(s, H) \in U$, the conclusion of 1.1 holds for $Z(s)$ and H . This parameterized variant of 1.1 will be used implicitly in the proof of 1.4.

I conjecture more generally:

Conjecture 1.3.- Let Z be smooth and connected of dimension n , let $f : Z \rightarrow \mathbb{P}^N$ be a morphism with finite fibers, let c be an integer, and let L be a linear subspace of $\mathbb{P}^N$ of

codimension c , with $n^* = n - c$. Choose a Riemannian metric on $\mathbb{P}^N$, and let $V(\varepsilon)$ be the set of points p of $\mathbb{P}^N$ such that $d(p, L) < \varepsilon$. Then, for ε sufficiently small, the natural map

$$\pi_i(f^{-1}(V(\varepsilon))) \rightarrow \pi_i(Z)$$

is bijective for $i < n^*$ and surjective for $i = n^*$.

For general L , the inclusion of $f^{-1}(L)$ into $f^{-1}(V(\varepsilon))$ is, for ε sufficiently small, a homotopy equivalence: 1.3 is indeed a generalization of 1.1.

If f is no longer assumed quasi-finite, and if the set of points of $\mathbb{P}^N$ where the fiber of f has dimension k has dimension $\varphi(k)$, then n^* should be redefined as

$$n^* = 2 \dim Z - \sup(2k + \varphi(k) + \inf(\varphi(k), c - 1)) - 1.$$

Conjecture 1.3 should moreover follow from a more precise statement saying that Z is obtained from $f^{-1}(V(\varepsilon))$ by attaching handles that kill spheres of dimension $\geq n^*$.

We shall be content with weaker results than 1.3. We deduce them from the special case of 1.1 in which Z is the complement of a hypersurface.

Lemma 1.4.- Let Z be normal and connected, let $f : Z \rightarrow \mathbb{P}^N$ be a morphism, let n be the dimension of $f(Z)$, and let c be an integer. Assume $n - c \geq 1$. If L is a general linear subspace of $\mathbb{P}^N$ of codimension c , then $f^{-1}(L)$ is connected and, for $z \in f^{-1}(L)$, the natural map from $\pi_1(f^{-1}(L), z)$ to $\pi_1(Z, z)$ is surjective.

One may replace Z by a nonempty Zariski-open subset Z' : for general L , $f^{-1}(L)$ is still normal, and Z' (respectively $Z' \cap f^{-1}(L)$) is obtained from Z (respectively $f^{-1}(L)$) by removing an algebraic subset with empty interior. By normality, this does not change π_0 , can only enlarge π_1 , and one considers the commutative diagram

$$\begin{array}{ccc} \pi_1(Z' \cap f^{-1}(L), z) & \twoheadrightarrow & \pi_1(f^{-1}(L), z) \\ \downarrow \downarrow & & \\ \pi_1(Z', z) & \twoheadrightarrow & \pi_1(Z, z). \end{array}$$

In particular, we may and do assume that Z is smooth. A reader interested only in the proof of Theorem 1 could in fact have assumed this from the outset; throughout the remainder of the section, that reader may replace “normal” by “smooth.”

After shrinking Z , we may and do assume that $f(Z)$ is locally closed and that Z is a topologically locally trivial fiber space over $f(Z)$.

If $c = 1$ and f is dominant, then $f(Z)$ is a Zariski-open subset of $\mathbb{P}^N$ - after shrinking Z , we may assume it is the complement of a hypersurface - and it remains only to apply 1.1 to $f(Z)$ and compare the long exact homotopy sequences for $Z \rightarrow f(Z)$ and $f^{-1}(L) \rightarrow L \cap f(Z)$.

Continue to assume $c = 1$. If f is not dominant, let A be a

linear subspace of dimension $N - n - 1$, let B be the projective space of linear subspaces of dimension $N - n$ containing A , and let $q : \mathbb{P}^N - A \rightarrow B$ send x to the subspace containing it. For A disjoint from $\overline{f(Z)}$, the restriction of q to $\overline{f(Z)}$ is a finite morphism, hence dominant because $\dim \overline{f(Z)} = \dim B$, and $qf : Z \rightarrow B$ is dominant. Applying the lemma to qf with $c = 1$ gives the conclusion of the lemma for a general hyperplane L among those containing A . Since A is only required to lie in a dense open subset of a suitable Grassmannian, the conclusion of the lemma also holds for general L (cf. 1.2).

The proof is completed by induction on c : a general linear subspace of codimension $c - 1$ in a general hyperplane is a general linear subspace of codimension c (cf. 1.2).

Lemma 1.5.- Let Z , f , n , and c be as in 1.4. Every linear subspace L of $\mathbb{P}^N$ of codimension c has a fundamental system of open neighborhoods V such that $f^{-1}(V)$ is connected and, for $z \in f^{-1}(V)$, the natural map from $\pi_1(f^{-1}(V), z)$ to $\pi_1(Z, z)$ is surjective.

Let Gr be the Grassmannian of linear subspaces of $\mathbb{P}^N$ of codimension c , let $Y \subset Z \times \text{Gr}$ be the set of pairs (z, L) such that $f(z) \in L$, and let G be a nonempty Zariski-open subset of Gr such that over G , Y is a topologically locally trivial fiber space. The conclusion of 1.4 then holds for $L \in G$ (cf. 1.2).

For every open subset U of Gr , write U^* for the open subset of $\mathbb{P}^N$ that is the union of the $L \in U$. We shall prove that if U is connected, the conclusion of 1.5 holds for $V = U^*$. The lemma will follow.

Let us prove that if U is connected, $f^{-1}(U^*)$ is connected as well. The intersection $U' = U \cap G$ is connected, since it is obtained from U by removing a triangulable subset of real codimension ≥ 2 . By 1.4, the union $f^{-1}(U'^*)$ of the $f^{-1}(L)$ for $L \in U'$ is therefore connected. We shall show that it is dense in $f^{-1}(U^*)$.

The image $f(Z)$ of Z is not contained in the complement F of G^* : since $n - c \geq 0$, the $L \in G$ meet $f(Z)$. If $z \in Z$, the image of any neighborhood of z is not contained in F ; otherwise the image of Z would be contained in F by analytic continuation. If $z \in f^{-1}(U^*)$, there are therefore points z' arbitrarily near z such that $f(z') \in G^*$. In the set of $L \in \text{Gr}$ containing z' , the trace of G is dense, since it is nonempty. If $z \in f^{-1}(L)$, with $L \in U$, this makes it possible to find z' as above lying in some $f^{-1}(L')$, with L' in G arbitrarily close to L . In particular one may take $L' \in U'$, in which case $z' \in f^{-1}(U'^*)$. This proves the density of $f^{-1}(U'^*)$ in $f^{-1}(U^*)$ and the connectedness of $f^{-1}(U^*)$.

Finally let $L \in U'$. We have $f^{-1}(L) \subset f^{-1}(U^*) \subset Z$. By 1.4, $\pi_1(f^{-1}(L))$ maps onto $\pi_1(Z)$. A fortiori the same is true for $\pi_1(f^{-1}(U^*))$.

We shall deduce from this a variant of Theorem 3.

THEOREM 1.6.- Let $P = (\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P , let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be normal and connected, let $h : Z \rightarrow P$, let $n = \dim h(Z)$, and let $c = (r - 1)m$ be the codimension of Δ in P . If $n - c \geq 1$, then Δ has a fundamental system of open neighborhoods V such that $h^{-1}(V)$ is connected. If $h^{-1}(V)$ is connected and $z \in h^{-1}(V)$, the natural map from $\pi_1(h^{-1}(V), z)$ to $\pi_1(V, z)$ is surjective.

As in 1.4, one may shrink Z , and in particular assume that Z is smooth. We do so. Let $(H_\alpha)_0 \leq \alpha \leq m$ be a family of hyperplanes in $\mathbb{P}^m$ such that

- a) the intersection of the H_α is empty: in suitable projective coordinates (X_α) , they are given by $X_\alpha = 0$;
- b) $h(Z)$ is not contained in any $\text{pr}_{-i}^{-1}(H_\alpha)$. After shrinking Z , we may and do assume that $\overline{h(Z)}$ is disjoint from the $\text{pr}_{-i}^{-1}(H_\alpha)$;
- c) $\overline{h(Z)} \cap \Delta$ is not contained in the union of the $\delta(H_\alpha)$: since Δ belongs to a connected family of cycles filling all of P , one has $\overline{h(Z)} \cap \Delta \neq \emptyset$, and it is possible to choose H_α satisfying c).

Let $A_\alpha = \mathbb{P}^m - H_\alpha$, let $B_\alpha = A_\alpha^r \subset P$, and let $\Delta_\alpha = \delta(A_\alpha)$ be the diagonal of B_α . The spaces A_α and B_α are affine spaces and Δ_α is an affine subspace of B_α . The projective completion of A_α is $\mathbb{P}^m$. Denote by $\bar{B}_\alpha$ the projective completion of B_α , and by $\bar{\Delta}_\alpha \subset \bar{B}_\alpha$ the completion of Δ_α .

Fix α , and write simply $H, A, B, \Delta_A, \bar{B}, \bar{\Delta}_A$ for $H_\alpha, A_\alpha, B_\alpha, \Delta_\alpha, \bar{B}_\alpha$, and $\bar{\Delta}_\alpha$.

Lemma 1.7.- For every neighborhood U of $\Delta \subset P$, there exists a neighborhood V of $\bar{\Delta}_A$ in $\bar{B}$ such that $V \cap B \subset U \cap B$.

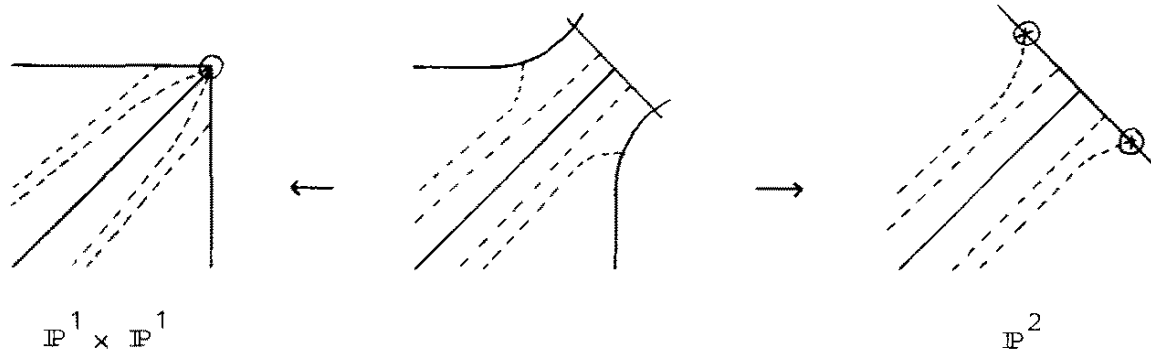
Choose on A the structure of a Hermitian vector space compatible with its affine structure. A fundamental system of neighborhoods U of Δ (respectively V of $\bar{\Delta}_A$) then has traces on B of the following form:

$U \cap B = \{(x_i) \mid \text{If } |x_i| \leq R \text{ for some } i, \text{ then } \forall j \forall k |x_j - x_k| < \varepsilon; \text{ if the } |x_i| \text{ are } \geq R, \text{ then for all } j, k \text{ the angle between the complex lines generated by } x_j \text{ and } x_k \text{ is } < \varepsilon\}$

$V \cap B = \{(x_i) \mid \text{If } |x_i| \leq R \text{ for some } i, \text{ then } \forall j \forall k |x_j - x_k| < \varepsilon; \text{ if the } |x_i| \text{ are } \geq R, \text{ then } \forall j \forall k |x_j - x_k| \leq \varepsilon \inf\{|x_i|\}\}$

The lemma follows.

When $r = m = 1$, the two compactifications $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ of $\mathbb{A}^2$ are topped by a third as follows: the drawing represents the locus at infinity and the diagonal; the circled points are those that must be blown up to obtain the third compactification; U and V are indicated by the dotted lines:



A similar construction can be made for arbitrary r and m .

Return to the proof of 1.6. Apply 1.5 to $h : Z \rightarrow \bar{B}_\alpha$ and to Δ_α . It follows that Δ_α has an arbitrarily small open neighborhood V_α such that $h^{-1}(V_\alpha) = h^{-1}(V_\alpha \cap B_\alpha)$ is connected and its π_1 maps onto that of Z . By 1.7, whatever the neighborhood U of Δ in P , V_α can be chosen so that $V_\alpha \cap B_\alpha \subset U$.

The union V of the open subsets $V_\alpha \cap B_\alpha$ of P is an arbitrarily small neighborhood of Δ . Hypothesis b) on the H_α ensures that the intersection of the $V_\alpha \cap B_\alpha$ meets $h(Z)$. Hence $h^{-1}(V)$, a union of connected open subsets $h^{-1}(V_\alpha)$ with nonempty common intersection, is connected. Since the π_1 of one of the $h^{-1}(V_\alpha)$ maps onto that of Z , a fortiori the same is true for $h^{-1}(V)$, and this completes the proof.

2. Proof of Theorem 1

2.1.- Let $C \subset \mathbb{P}^2$ be a reduced plane curve, let D be an irreducible component of C , let $C' = \overline{(C - D)}$ be the union of the other components, let $\text{Sing}(C)$ be the set of singular points of C , and let $D^* = D - \text{Sing}(C)$. Choose a base point $O \in \mathbb{P}^2 - C$, and let γ be a loop in $\mathbb{P}^2 - C$ of the following type: γ goes from O along a path ch to a point p' near a point p of D^* , goes once around D^* in the positive direction, and returns to O along ch^{-1} . Since D^* is connected, all loops of this type are conjugate. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the normal subgroup they generate is $\pi_1(\mathbb{P}^2 - C')$.

PROPOSITION 2.2.- Suppose that the singular points of C lying on D are double points with distinct tangents. Then γ is central in $\pi_1(\mathbb{P}^2 - C, O)$.

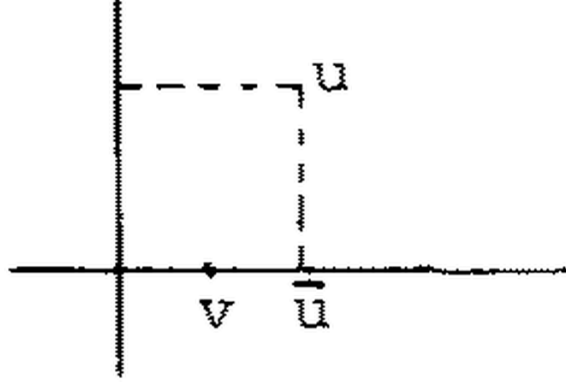
Let j be the inclusion of $\mathbb{P}^2 - C$ into $\mathbb{P}^2$, let $g : \tilde{D} \rightarrow D$ be the normalization of D , and apply 1.6 to

$$h = j \times g : (\mathbb{P}^2 - C) \times \tilde{D} \rightarrow \mathbb{P}^2 \times \mathbb{P}^2.$$

To describe conveniently the inverse images of certain neighborhoods of the diagonal, choose on $\mathbb{P}^2$ a Riemannian structure such that, near each

point $P \in D \cap \text{Sing}(C)$, $\mathbb{P}^2$ is Euclidean and the two branches of C through P are orthogonal real 2-planes.

If $(u, v) \in (\mathbb{P}^2 - C) \times \widetilde{D}$ lies in the inverse image of a sufficiently small neighborhood of the diagonal, that is, if u and v are sufficiently close, then u is near D and one can drop a short perpendicular from u to D . When u is near $\text{Sing}(D)$, there are even two such perpendiculars, one for each branch. Choose the one whose foot $\bar{u}$ is close to v on $\widetilde{D}$:



The point $\bar{u}$ lies in D^* .

Conversely, let $\varepsilon > 0$ be sufficiently small, and let $q : U^* \rightarrow D^*$ be the punctured-disk subbundle of the normal bundle of D^* in $\mathbb{P}^2$ consisting of nonzero normal vectors of length $< \varepsilon$. Let E be the set of pairs (x, v) with $x \in U^*$, $v \in \widetilde{D}$, and $d(q(x), v) < \varepsilon$ (distance measured on $\widetilde{D}$). The map $\text{pr}_1 : E \rightarrow U^*$ is a homotopy equivalence and has the space of pairs $(x, q(x))$ as a section. The map $\text{exp} : (x, v) \mapsto (\text{exp}(x), v)$ from E to $(\mathbb{P}^2 - C) \times \widetilde{D}$ identifies E with the inverse image of a neighborhood of Δ , and as $\varepsilon \rightarrow 0$ one obtains inverse images of a fundamental system of neighborhoods. Applying 1.6, one finds that for every $O \in U^*$, exp and q induce a surjection

$$(\text{exp}, q)^* : \pi_1(U^*, O) \twoheadrightarrow \pi_1(\mathbb{P}^2 - C, \text{exp}(O)) \times \pi_1(\widetilde{D}, q(O)).$$

There is an exact homotopy sequence

$$\mathbb{Z} \rightarrow \pi_1(U^*, O) \rightarrow \pi_1(D^*, q(O)) \rightarrow 1.$$

The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop going once around D^* in the positive direction. Since $(\text{exp}, q)^*$, and hence $\text{exp}^* : \pi_1(U^*, O) \rightarrow \pi_1(\mathbb{P}^2 - C, \text{exp}(O))$, is surjective, this loop also defines a central element of $\pi_1(\mathbb{P}^2 - C, \text{exp}(O))$. It is γ .

2.3 Proof of Theorem 1: For every irreducible component D of C , let $\gamma(D)$ be the corresponding central element of $\pi_1(\mathbb{P}^2 - C, O)$. The quotient of $\pi_1(\mathbb{P}^2 - C, O)$ by the - central - subgroup generated by the $\gamma(D)$ is $\pi_1(\mathbb{P}^2, O) = \{e\}$. The theorem follows.

BIBLIOGRAPHY

- [1] S. ABHYANKAR - Tame coverings and fundamental groups of algebraic varieties I, Amer. J. of Math., 81(1959), 46-94.
- [2] D. CHENIOT - Un théorème du type de Lefschetz, Annales de l'Institut Fourier, 25 1(1975), 195-213.
- [2 bis] D. CHENIOT - Une démonstration du théorème de Zariski sur les sections hyperplanes d'une hypersurface projective et du théorème de Van Kampen sur le groupe fondamental du complémentaire d'une courbe plane, Comp. Math., 27 2(1973), 141-158.
- [3] W. FULTON - On the fundamental group of the complement of a node curve, Ann. of Math., 111 2(1980), 407-409.
- [4] W. FULTON and J. HANSEN - A connectedness theorem for projective varieties, with applications to intersections and singularities of mappings, Ann. of Math., 109 1(1979), 159-166.
- [5] T. GAFFNEY and R. LAZARSFELD - On the ramification of branched coverings of $\mathbb{P}^n$, Inv. Math., 59 1(1980), 53-58.
- [6] H. HAMM et LÊ DŨNG TRANG - Un théorème de Zariski du type de Lefschetz, Ann. Sci. E.N.S., 6 3(1973), 317-366.
- [7] J.-P. SERRE - Revêtements ramifiés du plan projectif (d'après S. Abhyankar), Sém. Bourbaki 204, mai 1960.
- [8] O. ZARISKI - On the problem of existence of algebraic functions of two variables possessing a given branch curve, Amer. J. of Math., 51(1929), 305-328.
- [9] O. ZARISKI - Algebraic surfaces, second supplemented edition, Ergebnisse 61, Springer-Verlag.

I have just received (1/10/80) a very fine account surveying the methods studied here, their variants, and their many applications: W. FULTON and R. LAZARSFELD - Connectivity and its applications in algebraic geometry.

Pierre DELIGNE
 Institut des Hautes Etudes
 Scientifiques
 35 route de Chartres
 F-91440 BURES-SUR-YVETTE